Arshia Riaz

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SUMMARY

Data Science and Healthcare Technology professional with experience building AI-enabled products, behavioral data analysis pipelines, and end-to-end ML applications. Skilled in Python, R, statistical modeling, and Google Cloud, with a background in healthcare consulting, EHR systems, cognitive and physiological research, and applied neuroscience. Published author in peer-reviewed medical journals.

EDUCATION

The University of Texas at Austin, Austin, TX — May 2025
Master of Science, Data Science, GPA 3.67

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Data Exploration & Visualization, Predictive Modeling, Data Science in Healthcare, Data Structures & Algorithms

The University of Texas at Austin, Austin, TX — May 2023
Bachelor of Science, Neuroscience, GPA 3.88

- **Relevant Coursework:** Neural Systems I, II, & III, Programming/Data Analysis in Modern Neuroscience, Neural Computation, Behavioral Neuroscience, Sensorimotor Systems, Cognitive Psychology, Human Movement Science

CERTIFICATION

- Google Cloud Platform Professional Machine Learning Engineer — March 2026

SKILLS

- **Programming Languages:** Python, SQL, R, MATLAB
- **Data Science & Machine Learning:** Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Model Training & Evaluation, Feature Engineering, Statistical Modeling, Behavioral Data Analysis, Supervised Learning
- **Generative AI & LLMs:** Embeddings, Fine-Tuning, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Evaluation
- **Data Engineering & Analytics:** Data Cleaning, ETL Pipelines, API Development (Flask, FastAPI), BigQuery, Data Modeling
- **Cloud & MLOps:** Google Cloud Platform (BigQuery, Vertex AI), CI/CD, Git, Model Deployment, Firebase, Streamlit, Plotly
- **Domain Expertise:** Healthcare Data, EHR Systems (NextGen, DrChrono), Healthcare Technology, HIPAA, HL7, FHIR, Applied Neuroscience, Behavioral Analysis, Cognitive Performance Modeling, Human Movement & Biomechanics, Psychophysiology, Statistical Parametric Mapping (SPM)

APPLIED DATA & PRODUCT DEVELOPMENT EXPERIENCE

Psychedelic Neural Fingerprinting — Neuroimaging ML Analysis — November 2025 – March 2026

- Demo: https://psychedelic-neural-fingerprinting.streamlit.app
- Built end-to-end ML pipeline on fMRI neuroimaging data across two public datasets (N=22 subjects) to classify psychedelic-induced brain states; SVM classifier achieved AUC=0.967 under leave-one-subject-out cross-validation with permutation testing (p=0.001), preventing subject leakage.
- Identified pharmacologically specific connectivity signature: classifier trained on LSD generalized to psilocybin vs methylphenidate (AUC=0.918), with longitudinal analysis showing sustained post-psilocybin brain reorganization not observed after stimulant controls.

CareNav Health — AI-Enabled Healthcare Navigation Platform — August 2025 – Present

- Demo: https://carenav.vercel.app
- Built ML-backed healthcare decision system mapping symptoms to care and estimating costs using CPT-based procedure bundling and pricing model ($R^2 \approx 0.92$, MAE≈$17).
- Designed data pipeline and RAG-based benefits system integrating insurance extraction (OCR) and structured plan data to generate personalized, grounded cost and coverage insights.

Amalma — Maternal Health Data Tracking App — July 2025 – September 2025

- Demo: https://amalma-app.web.app

- Built a HIPAA-conscious maternal health tracking app with Firebase authentication, Firestore cloud database, and mobile-optimized UI, enabling real-time pumping/feeding logs and analytics. Designed with scalable cloud architecture for integration with future AI-driven health insights.

Cognitive Fatigue Analyzer — Behavioral Decision Fatigue Modeling App June 2025 – July 2025

- Demo: https://cognitive-fatigue-analyzer.streamlit.app
- Built and deployed an interactive behavioral data science application modeling cognitive decision fatigue from physiological and behavioral signals across 25,000 decision states; conducted full statistical analysis surfacing a 59x error rate inflection at critical fatigue thresholds.
- Engineered 4 derived behavioral features grounded in cognitive load research; trained Random Forest classifier (96.7% accuracy, 5-fold CV) and regressor (R^2=0.998). Deployed Streamlit app with real-time fatigue prediction, recovery scenario modeling, and forward trajectory projection.

WORK EXPERIENCE

NextGen Healthcare

Specialist I, Professional Services Consulting July 2023 – June 2025

- Delivered consulting on best practices and workflows to 10+ enterprise healthcare clients, leading discovery sessions and configuring NextGen EHR systems to improve adoption and operational efficiency.
- Led 20+ discovery and training sessions to analyze workflows, gather requirements, and improve system adoption.
- Configured and customized EHR workflows, templates, and forms, contributing to faster onboarding and improved operational efficiency for clinical staff.
- Served as subject matter expert for the Ophthalmology Suite, identifying and documenting 10+ product enhancement opportunities for engineering teams.

Brightly Software

Client Success Intern May 2022 – December 2022

- Supported onboarding and troubleshooting for 15+ client accounts using Brightly enterprise software solutions.
- Assisted with solution configuration and issue resolution, helping improve client adoption and satisfaction.

The Optometry Center for Vision Therapy, Austin, TX

Vision Therapist October 2021 – May 2022

- Delivered individualized vision therapy sessions for 30+ patients, collaborating with optometrists to track progress and adjust treatment plans.

Center for Neuromotor and Biomechanics Research, Houston, TX

Research Assistant May 2021 – August 2021

- Conducted neuromotor adaptation research analyzing behavioral and physiological responses to simulated hypo-gravity conditions using the AlterG anti-gravity treadmill.
- Collected and processed experimental data for 20+ study sessions using MATLAB; contributed to analysis of lower-limb kinematic and EMG hysteresis effects across simulated gravitational unloading conditions, with findings published in Frontiers in Physiology (2023).

Houston Oculoplastic Associates, Houston, TX

Ophthalmic Technician January 2020 – August 2021

- Managed medical documentation in EHR systems (NextGen, DrChrono) and performed diagnostic vision tests (Visual Field, Eyelid Field) for 100+ patients.

PUBLICATIONS

- Malaya, C.A., et al. (2023). *Effects of simulated hypo-gravity on lower limb kinematic and electromyographic variables during anti-gravitational treadmill walking.* Frontiers in physiology, 14.
- Riaz, A., et al. (2022). *Epidemiological Aspects of Giant Cell Arteritis*. Giant-Cell Arteritis, IntechOpen. (Book Chapter)